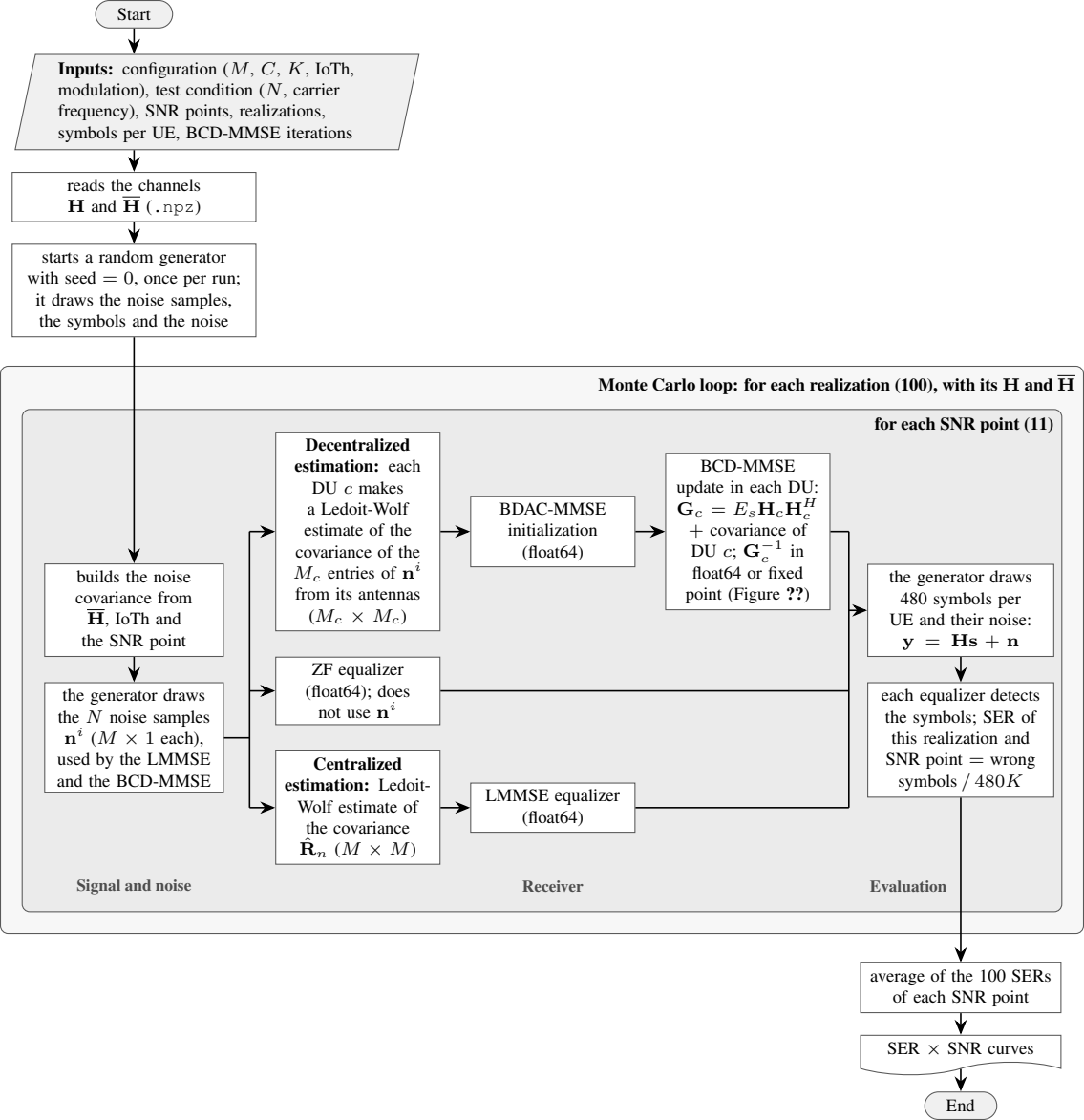

Start
Inputs: configuration ($M$, $C$, $K$, IoTh, modulation), test condition ($N$, carrier frequency), SNR points, realizations, symbols per UE, BCD-MMSE iterations
reads the channels $\mathbf{H}$ and $\overline{\mathbf{H}}$ (.npz)
starts a random generator with seed = 0, once per run; it draws the noise samples, the symbols and the noise
Monte Carlo loop: for each realization (100), with its $\mathbf{H}$ and $\overline{\mathbf{H}}$
for each SNR point (11)
builds the noise covariance from $\overline{\mathbf{H}}$, IoTh and the SNR point
the generator draws the $N$ noise samples $\mathbf{n}^i$ ($M \times 1$ each), used by the LMMSE and the BCD-MMSE
Decentralized estimation: each DU $c$ makes a Ledoit-Wolf estimate of the covariance of the $M_c$ entries of $\mathbf{n}^i$ from its antennas ($M_c \times M_c$)
ZF equalizer (float64); does not use $\mathbf{n}^i$
Centralized estimation: Ledoit-Wolf estimate of the covariance $\hat{\mathbf{R}}_n$ ($M \times M$)
BDAC-MMSE initialization (float64)
LMMSE equalizer (float64)
BCD-MMSE update in each DU: $\mathbf{G}_c = E_s \mathbf{H}_c \mathbf{H}_c^H$ + covariance of DU $c$; $\mathbf{G}_c^{-1}$ in float64 or fixed point (Figure ??)
the generator draws 480 symbols per UE and their noise: $\mathbf{y} = \mathbf{H}\mathbf{s} + \mathbf{n}$
each equalizer detects the symbols; SER of this realization and SNR point = wrong symbols / $480K$
Signal and noise
Receiver
Evaluation
average of the 100 SERs of each SNR point
SER × SNR curves
End